

It is not difficult to prove that a graph of minimum degree  $\delta$  and girth  $g$  has at least  $n_0(\delta, g)$  vertices (Exercise 7). Interestingly, one can obtain the same bound for its average degree:

**Theorem 1.3.4.** (Alon, Hoory & Linial 2002)

Let  $G$  be a graph. If  $d(G) \geq d \geq 2$  and  $g(G) \geq g \in \mathbb{N}$  then  $|G| \geq n_0(d, g)$ .

One aspect of Theorem 1.3.4 is that it guarantees the existence of a short cycle compared with  $|G|$ . Using just the easy minimum degree version of Exercise 7, we get the following rather general bound:

[2.3.1] **Corollary 1.3.5.** If  $\delta(G) \geq 3$  then  $g(G) < 2 \log |G|$ .

*Proof.* If  $g := g(G)$  is even then

$$n_0(3, g) = 2 \frac{2^{g/2} - 1}{2 - 1} = 2^{g/2} + (2^{g/2} - 2) > 2^{g/2},$$

while if  $g$  is odd then

$$n_0(3, g) = 1 + 3 \frac{2^{(g-1)/2} - 1}{2 - 1} = \frac{3}{\sqrt{2}} 2^{g/2} - 2 > 2^{g/2}.$$

As  $|G| \geq n_0(3, g)$ , the result follows.  $\square$

walk

A *walk* (of length  $k$ ) in a graph  $G$  is a non-empty alternating sequence  $v_0 e_0 v_1 e_1 \dots e_{k-1} v_k$  of vertices and edges in  $G$  such that  $e_i = \{v_i, v_{i+1}\}$  for all  $i < k$ . If  $v_0 = v_k$ , the walk is *closed*. If the vertices in a walk are all distinct, it defines an obvious path in  $G$ . In general, every walk between two vertices contains<sup>4</sup> a path between these vertices (proof?).

## 1.4 Connectivity

connected

A graph  $G$  is called *connected* if it is non-empty and any two of its vertices are linked by a path in  $G$ . If  $U \subseteq V(G)$  and  $G[U]$  is connected, we also call  $U$  itself connected (in  $G$ ). Instead of ‘not connected’ we usually say ‘disconnected’.

[1.5.2] **Proposition 1.4.1.** The vertices of a connected graph  $G$  can always be enumerated, say as  $v_1, \dots, v_n$ , so that  $G_i := G[v_1, \dots, v_i]$  is connected for every  $i$ .

---

<sup>4</sup> We shall often use terms defined for graphs also for walks, as long as their meaning is obvious.

*Proof.* Pick any vertex as  $v_1$ , and assume inductively that  $v_1, \dots, v_i$  have been chosen for some  $i < |G|$ . Now pick a vertex  $v \in G - G_i$ . As  $G$  is connected, it contains a  $v-v_1$  path  $P$ . Choose as  $v_{i+1}$  the last vertex of  $P$  in  $G - G_i$ ; then  $v_{i+1}$  has a neighbour in  $G_i$ . The connectedness of every  $G_i$  follows by induction on  $i$ .  $\square$

Let  $G = (V, E)$  be a graph. A maximal connected subgraph of  $G$  is a *component* of  $G$ . Clearly, the components are induced subgraphs, and their vertex sets partition  $V$ . Since connected graphs are non-empty, the empty graph has no components.

component

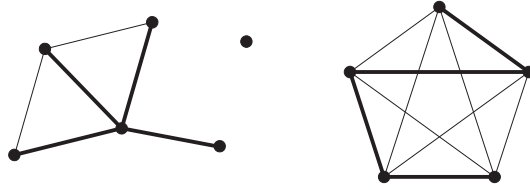


Fig. 1.4.1. A graph with three components, and a minimal spanning connected subgraph in each component

If  $A, B \subseteq V$  and  $X \subseteq V \cup E$  are such that every  $A$ - $B$  path in  $G$  contains a vertex or an edge from  $X$ , we say that  $X$  *separates* the sets  $A$  and  $B$  in  $G$ . Note that this implies  $A \cap B \subseteq X$ . We say that  $X$  *separates* two vertices  $a, b$  if it separates the sets  $\{a\}, \{b\}$  but  $a, b \notin X$ . The set  $X$  *separates*  $G$ , and is a *separator* in or of  $G$ , if  $X$  separates some two vertices in  $G$ . Separating sets of edges have no generic name, but some such sets do; see Section 1.9 for the definition of *cuts* and *bonds*. A vertex which separates two other vertices of the same component is a *cutvertex*, and an edge separating its ends is a *bridge*. Thus, the bridges in a graph are precisely those edges that do not lie on any cycle.

separate

separator

cutvertex

bridge

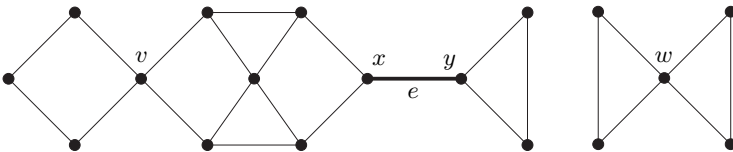


Fig. 1.4.2. A graph with cutvertices  $v, x, y, w$  and bridge  $e = xy$

The unordered pair  $\{A, B\}$  is a *separation* of  $G$  if  $A \cup B = V$  and  $G$  has no edge between  $A \setminus B$  and  $B \setminus A$ . Clearly, the latter is equivalent to saying that  $A \cap B$  separates  $A$  from  $B$ . If both  $A \setminus B$  and  $B \setminus A$  are non-empty, the separation is *proper*. The number  $|A \cap B|$  is the *order* of the separation  $\{A, B\}$ ; the sets  $A, B$  are its *sides*.

separation

$G$  is called  *$k$ -connected* (for  $k \in \mathbb{N}$ ) if  $|G| > k$  and  $G - X$  is connected for every set  $X \subseteq V$  with  $|X| < k$ . In other words, no two vertices of  $G$

 $k$ -connected

connectivity  
 $\kappa(G)$

$\ell$ -edge-  
connected  
edge-  
connectivity  
 $\lambda(G)$

are separated by fewer than  $k$  other vertices. Every (non-empty) graph is 0-connected, and the 1-connected graphs are precisely the non-trivial connected graphs. The greatest integer  $k$  such that  $G$  is  $k$ -connected is the *connectivity*  $\kappa(G)$  of  $G$ . Thus,  $\kappa(G) = 0$  if and only if  $G$  is disconnected or a  $K^1$ , and  $\kappa(K^n) = n - 1$  for all  $n \geq 1$ .

If  $|G| > 1$  and  $G - F$  is connected for every set  $F \subseteq E$  of fewer than  $\ell$  edges, then  $G$  is called  *$\ell$ -edge-connected*. The greatest integer  $\ell$  such that  $G$  is  $\ell$ -edge-connected is the *edge-connectivity*  $\lambda(G)$  of  $G$ . In particular, we have  $\lambda(G) = 0$  if  $G$  is disconnected.

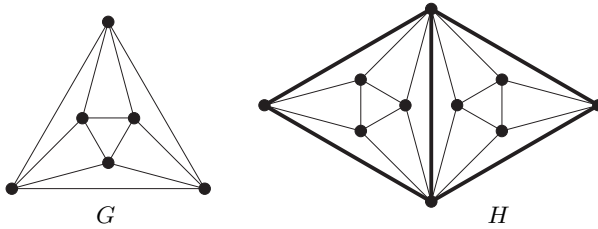


Fig. 1.4.3. The octahedron  $G$  (left) with  $\kappa(G) = \lambda(G) = 4$ , and a graph  $H$  with  $\kappa(H) = 2$  but  $\lambda(H) = 4$

[3.2.1] **Proposition 1.4.2.** *If  $G$  is non-trivial then  $\kappa(G) \leq \lambda(G) \leq \delta(G)$ .*

*Proof.* The second inequality follows from the fact that all the edges incident with a fixed vertex separate  $G$ . To prove the first, let  $F$  be a set of  $\lambda(G)$  edges such that  $G - F$  is disconnected. Such a set exists by definition of  $\lambda$ ; note that  $F$  is a minimal separating set of edges in  $G$ . We show that  $\kappa(G) \leq |F|$ .

Suppose first that  $G$  has a vertex  $v$  that is not incident with an edge in  $F$ . Let  $C$  be the component of  $G - F$  containing  $v$ . Then the vertices of  $C$  that are incident with an edge in  $F$  separate  $v$  from  $G - C$ . Since no edge in  $F$  has both ends in  $C$  (by the minimality of  $F$ ), there are at most  $|F|$  such vertices, giving  $\kappa(G) \leq |F|$  as desired.

Suppose now that every vertex is incident with an edge in  $F$ . Let  $v$  be any vertex, and let  $C$  be the component of  $G - F$  containing  $v$ . Then the neighbours  $w$  of  $v$  with  $vw \notin F$  lie in  $C$  and are incident with distinct edges in  $F$  (distinct by the minimality of  $F$ , as earlier), giving  $d_G(v) \leq |F|$ . As  $N_G(v)$  separates  $v$  from any other vertices in  $G$ , this yields  $\kappa(G) \leq |F|$  – unless there are no other vertices, i.e. unless  $\{v\} \cup N(v) = V$ . But  $v$  was an arbitrary vertex. So we may assume that  $G$  is complete, giving  $\kappa(G) = \lambda(G) = |G| - 1$ .  $\square$

By Proposition 1.4.2, high connectivity requires a large minimum degree. Conversely, large minimum degree does not ensure high connectivity, not even high edge-connectivity (examples?). It does, however, imply the existence of a highly connected subgraph:

**Theorem 1.4.3.** (Mader 1972)

[7.2.3]  
[11.2.3]

Let  $0 \neq k \in \mathbb{N}$ . Every graph  $G$  with  $d(G) \geq 4k$  has a  $k$ -connected subgraph. In fact, every such  $G$  has a  $(k+1)$ -connected subgraph  $H$  such that  $d(H) > d(G) - 2k \geq 2k$ .

*Proof.* Put  $\gamma := \varepsilon(G) (\geq 2k)$ , and consider the subgraphs  $G' \subseteq G$  such that

(1.2.2)  
(1.3.1)  
 $\gamma$

$$|G'| \geq 2k \quad \text{and} \quad \|G'\| > \gamma(|G'| - k). \quad (*)$$

Such graphs  $G'$  exist since  $G$  is one; let  $H$  be one of smallest order.

$H$

No graph  $G'$  as in  $(*)$  can have order exactly  $2k$ , since this would imply that  $\|G'\| > \gamma k \geq 2k^2 > \binom{|G'|}{2}$ . The minimality of  $H$  therefore implies that  $\delta(H) > \gamma$ : otherwise we could delete a vertex of degree at most  $\gamma$  and obtain a graph  $G' \subsetneq H$  still satisfying  $(*)$ . In particular, we have  $|H| \geq \gamma$ . Dividing the inequality of  $\|H\| > \gamma|H| - \gamma k$  from  $(*)$  by  $|H|$  therefore yields  $\varepsilon(H) > \gamma - k$ , as desired.

It remains to show that  $H$  is  $(k+1)$ -connected. If not, then  $H$  has a proper separation  $\{U_1, U_2\}$  of order at most  $k$ ; put  $H[U_i] =: H_i$ . Since any vertex  $v \in U_1 \setminus U_2$  has all its  $d(v) \geq \delta(H) > \gamma$  neighbours from  $H$  in  $H_1$ , we have  $|H_1| \geq \gamma \geq 2k$ . Similarly,  $|H_2| \geq 2k$ . As by the minimality of  $H$  neither  $H_1$  nor  $H_2$  satisfies  $(*)$ , we thus have

$H_1, H_2$

$$\|H_i\| \leq \gamma(|H_i| - k)$$

for  $i = 1, 2$ . But then

$$\begin{aligned} \|H\| &\leq \|H_1\| + \|H_2\| \\ &\leq \gamma(|H_1| + |H_2| - 2k) \\ &\leq \gamma(|H| - k) \quad (\text{as } |H_1 \cap H_2| \leq k), \end{aligned}$$

which contradicts  $(*)$  for  $H$ .  $\square$

## 1.5 Trees and forests

An *acyclic* graph, one not containing any cycles, is called a *forest*. A connected forest is called a *tree*. (Thus, a forest is a graph whose components are trees.) The vertices of degree 1 in a tree are its *leaves*,<sup>5</sup> the others are its *inner vertices*. Every non-trivial tree has a leaf – consider, for example, the ends of a longest path. This little fact often comes in handy, especially in induction proofs about trees: if we remove a leaf from a tree, what remains is still a tree.

forest  
tree  
leaf

<sup>5</sup> ... except that the *root* of a tree (see below) is never called a leaf, even if it has degree 1.